

EXP NO: 01

DATE:
17/07/2026

VERIFICATION OF GATES

Aim:-

To study and verify the truth table of logic gates.

Apparatus Required:

All the basic gates mentioned in the fig.

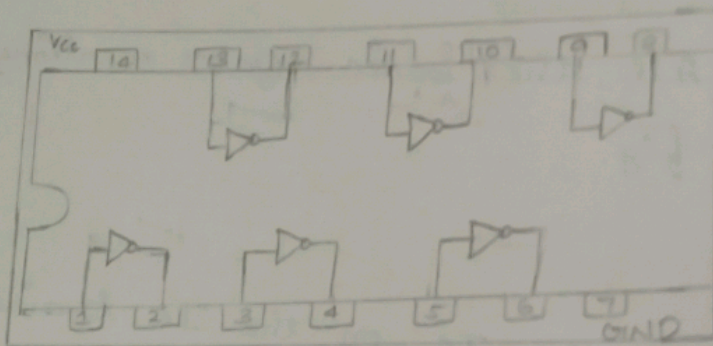
Procedure

1. Place the IC on the IC trainer kit.
2. Connect VCC and ground to respective pins of IC trainer kit.
3. Connect the inputs to the input switches provided in the IC Trainer kit.
4. Connect the outputs to the switches of O/P LEDs.
5. Apply various combinations of inputs according to the truth table and observe the conditions.

of LEDs

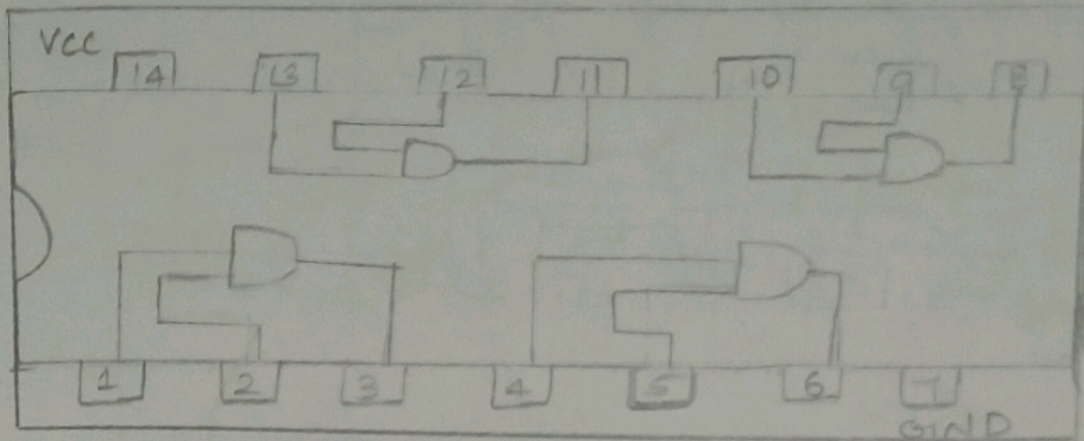
6. Disconnect output from the LEDs and note down the corresponding multimeter voltage readings for various combinations of inputs.

Inverter Gate (NOT Gate) - 7404LS



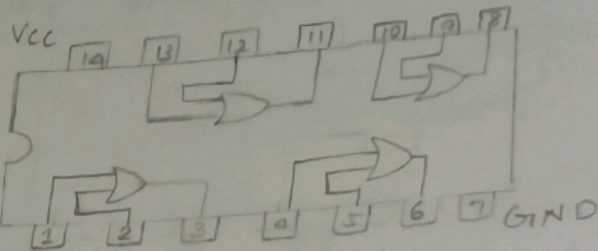
A	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)	Y ₅ (V)	Y ₆ (V)
0	1	5	5	5	5	5	5
1	0	0	0	0	0	0	0

2 - Input AND Gate - 7408LS



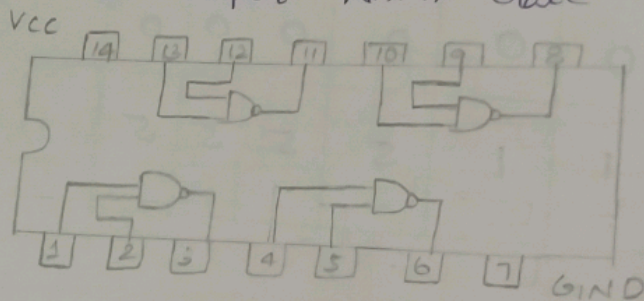
A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	0	0	0	0	0
0	1	0	0	0	0	0
1	0	0	0	0	0	0
1	1	1	5	5	5	5

2- Input OR Gate - 7432 LS



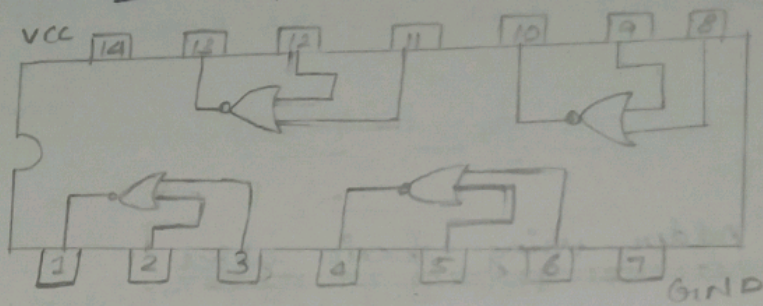
A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	0	0	0	0	0
0	1	1	5	5	5	5
1	0	1	5	5	5	5
1	1	1	5	5	5	5

2 - Input NAND Gate - 7400 LS



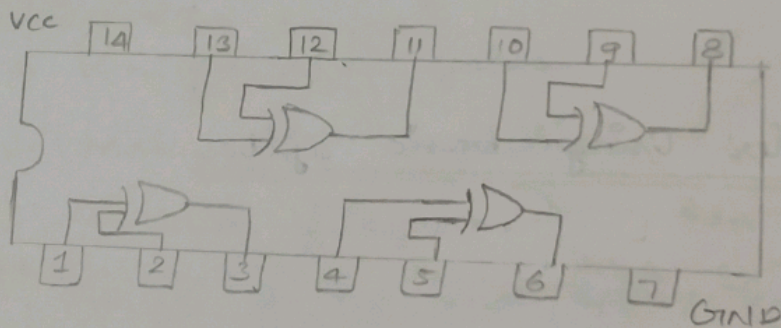
A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	1	5	5	5	5
0	1	1	5	5	5	5
1	0	1	5	5	5	5
1	1	0	0	0	0	0

2 - Input NOR Gate - 7402LS



A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	1	5	5	5	5
0	1	0	0	0	0	0
1	0	0	0	0	0	0
1	1	0	0	0	0	0

2 - Input Ex-OR Gate - 7486LS



A	B	O/P	Y ₁ (V)	Y ₂ (V)	Y ₃ (V)	Y ₄ (V)
0	0	0	0	0	0	0
0	1	1	5	5	5	5
1	0	1	5	5	5	5
1	1	0	5	5	5	5